Week 6 Final Report: Emergency Triage Classification

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Decision supported

I tested whether simple classifiers could support a triage nurse by estimating the recorded Emergency Severity Index (ESI) level from information available at triage. The models do not assign care independently.

After training-only tuning, logistic regression identified 7 of 16 ESI 1 visits, compared with 4 before tuning.

Dataset and split

The supplied dataset contains 55,121 emergency department visits and 225 usable columns after removal of the exported index. Only 77 visits are ESI 1, compared with 27,010 ESI 3 visits. I used 209 numeric inputs, including age, front-door vital signs and chief-complaint flags. Disposition and previousdispo were excluded because both are known after triage.

I held out 20% for the final test, stratified on ESI with random seed 42, then split the remaining training data again for tuning. The test set was not used to select settings. It contains 11,025 visits, including 16 ESI 1 cases.

Models and tuning

The required baselines are scaled logistic regression and an unscaled depth-5 decision tree, with a stratified dummy as the comparison floor. Following the interim feedback, I tested ESI 1 class weights of 2, 4, 8 and 12 for logistic regression. For the tree, I tested depths of 5, 8 and 12, minimum leaf sizes of 1, 5, 10 and 25, with and without balanced class weights.

I required validation ESI 1 recall of at least 0.25, then selected the eligible setting with the highest validation macro F1. This chose an ESI 1 weight of eight for logistic regression and a balanced depth-12 tree with a minimum leaf size of ten.

Held-out results

Model	Acc.	Macro F1	Weighted F1	ESI 1 recall	ESI 1 precision	Caught	False alerts
Stratified dummy	0.375	0.204	0.375	0.000	0.000	0/16	10
Logistic baseline	**0.683**	**0.508**	**0.677**	**0.250**	**0.500**	**4/16**	**4**
Tree baseline	0.547	0.207	0.448	0.000	0.000	0/16	0
Logistic tuned	**0.681**	**0.501**	**0.675**	**0.438**	**0.233**	**7/16**	**23**
Tree tuned	0.497	0.356	0.510	0.375	0.024	6/16	244

The tuned logistic model catches three additional ESI 1 visits while accuracy changes from 0.683 to 0.681. False ESI 1 alerts rise from 4 to 23. The tuned tree catches 6 cases but produces 244 false alerts, so I would not carry it forward.

Class-level results after tuning

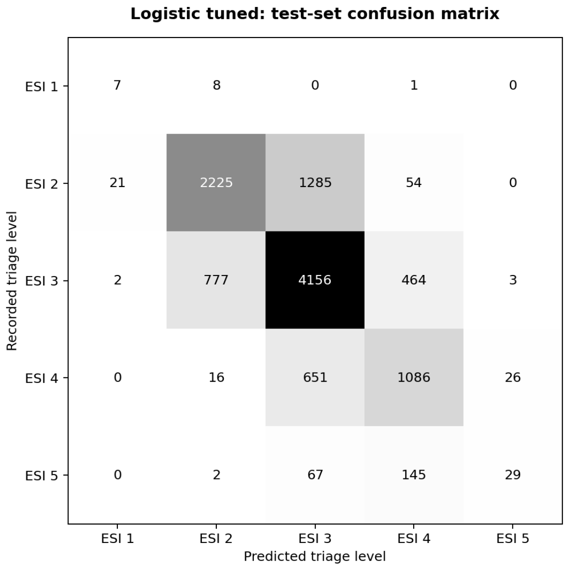
Logistic tuned

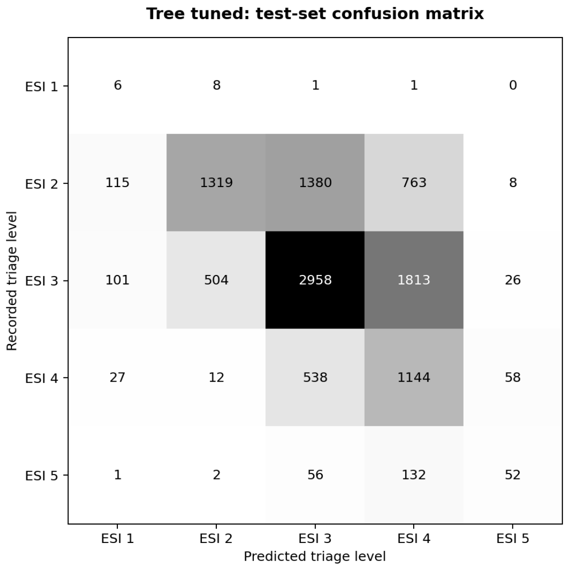
ESI	Precision	Recall	F1	Visits
1	0.233	0.438	0.304	16
2	0.735	0.621	0.673	3,585
3	0.675	0.769	0.719	5,402
4	0.621	0.610	0.615	1,779
5	0.500	0.119	0.193	243

Tree tuned

ESI	Precision	Recall	F1	Visits
1	0.024	0.375	0.045	16
2	0.715	0.368	0.486	3,585
3	0.600	0.548	0.572	5,402
4	0.297	0.643	0.406	1,779
5	0.361	0.214	0.269	243

Logistic regression keeps similar performance on ESI 2-5 while improving ESI 1 recall. The balanced tree spreads predictions across more classes, improving macro F1 over the depth-5 tree, but its ESI 1 precision falls to 0.024.





Rows are recorded ESI levels and columns are predicted levels. The tuned logistic model identifies 7 ESI 1 visits and misses 9; the tuned tree predicts ESI 1 too often.

ESI 1 error review

I reviewed the 16 held-out ESI 1 visits after evaluation using aggregate values only. Five of the seven caught cases had a stroke-alert flag. The nine misses were more mixed: four were grouped as other, two had shortness-of-breath flags, and single cases carried flags for hypotension, altered mental status, dizziness, a neurologic problem, poisoning or a mass. Multiple flags can appear for one visit.

Median triage measure	Caught, n=7	Missed, n=9
Age	73.0	81.0
Heart rate	81.0	81.0
Systolic BP	195.5	148.0
Diastolic BP	101.0	64.0
Respiratory rate	18.5	18.0
Oxygen saturation	98.0	96.0
Temperature	97.8	97.8
Glucose	118.0	113.0

The missed group is older at the median and has lower median oxygen saturation. The caught group's high blood-pressure medians are influenced by the stroke-alert cases. Sixteen visits are too few for a stable clinical conclusion, so a clinician should review the errors before new features are added.

Metric choice and decision

ESI 1 recall is the primary safety measure because it asks how many patients needing immediate intervention are recognised at that level. Accuracy and weighted F1 are still reported, but common ESI levels dominate both. The tuned logistic model improves ESI 1 recall by 0.188 without a material accuracy loss. It still misses 9 of 16 cases and raises the false-alert count, so this is a trade-off rather than a deployment result.

I would carry the tuned logistic model forward for probability calibration, clinician review and repeated stratified validation. Any threshold change should report critical cases caught alongside false alerts. External validation and subgroup checks are required before the model can support a real workflow.

Reproducibility and submitted artefacts

Item	Recorded value
Random seed	42
Split	80/20 held-out test; second stratified split inside training
Selected logistic	ESI 1 class weight = 8
Selected tree	max_depth=12; min_samples_leaf=10; class_weight=balanced
Notebook	notebooks/week6_final_baseline_models.ipynb
Metrics	w6_final_metrics.csv; w6_per_class_metrics.csv; w6_tuning_results.csv
ESI 1 review	w6_esi1_vital_profile.csv; w6_esi1_complaint_profile.csv
Explainer	docs/week6_clinical_explainer.mp4 and transcript

The full clinical CSV remains outside GitHub.